FATHIMA AFRAH M A
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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in data analysis, machine learning, NLP, computer vision, and agentic AI systems. Skilled in Python, C++, SQL, data preprocessing, statistical analysis, data visualization, and machine learning model evaluation. Experienced in building end-to-end ML pipelines, LLM-powered agent workflows, and structured data systems through internships and academic projects. Strong analytical and problem-solving abilities with attention to data accuracy, process improvement, and the delivery of reliable results.

TECHNICAL SKILLS

Programming & Data Analysis: Python, C++, SQL, Data Extraction, Data Preprocessing, Data Cleaning, Statistical Analysis, Exploratory Data Analysis (EDA)
Machine Learning & AI: Machine Learning, Deep Learning, NLP, Text Classification, Feature Engineering, Model Evaluation, Scikit-learn, TensorFlow, PyTorch, Keras, Logistic Regression
Agentic AI & LLMs: AI Agents, LLM-powered Workflows, Agentic Design, Retrieval-Augmented Responses, Prompt Engineering, Streamlit
Computer Vision: OpenCV, Image Processing, Image Preprocessing, Feature Extraction
Databases & Cloud: MySQL, SQLite, Firebase, AWS (S3, EC2 – Basic)
Data Visualization & Productivity: Power BI, Microsoft Excel, Data Visualization, Microsoft Office
Tools & Version Control: Git, GitHub, Jupyter Notebook
AI Tools: ChatGPT, Claude, GitHub Copilot

EXPERIENCE

Artificial Intelligence Intern — CodeClause (Remote) *Jul 2025 – Aug 2025*

- Performed data extraction, preprocessing, feature engineering, and model evaluation across end-to-end machine learning pipelines, achieving 85%+ model accuracy.
- Built and maintained structured data pipelines using SQLite and performed quality assurance to identify and resolve data inconsistencies.
- Applied machine learning and data analysis techniques to transform raw data into structured datasets suitable for model development and evaluation.
- Authored SOPs and process documentation to support consistent workflows and cross-team reporting.

PROJECTS

ParcelPilot AI Support Copilot | *Python, Streamlit, AI Agents, LLMs*

- Developed an AI-powered support copilot using Python and Streamlit to assist support teams with order/account investigation and customer queries.
- Implemented agentic workflows for order lookup, policy/document retrieval, customer-specific agreement handling, and evidence-based support decisions, generating source-referenced responses for reliable decision-making.
- Added investigation summaries and support escalation preparation, with explicit confirmation required before state-changing actions.
- Designed modular components for the application, including agent logic, data retrieval, and action handling.

Personality Prediction System | *Python, NLP, Scikit-learn, SQLite*

- Developed an NLP-based machine learning classification system to predict personality traits from text using tokenization, TF-IDF/vectorization, feature extraction, and Logistic Regression.
- Built a SQLite-backed data layer for structured storage and retrieval, and evaluated model performance using precision, recall, and F1-score.
- Performed data preprocessing and model evaluation to improve prediction reliability and maintain data quality throughout the pipeline.

SmartVein-ID: Infrared Hand Vein Authentication System | *Python, OpenCV, Raspberry Pi, SQLite*

- Built an end-to-end biometric authentication system using infrared imaging and vein-pattern recognition, integrating hardware (Raspberry Pi) with a software processing pipeline.
- Implemented image preprocessing techniques including noise reduction and edge detection to improve image quality and recognition.
- Maintained structured SQLite audit logs to support data integrity, traceability, and reliable system processing.

CERTIFICATIONS

- AWS Solutions Architecture Job Simulation — Amazon Web Services (Forage) | Jun 2025
- GenAI Powered Data Analytics Job Simulation — Tata Group (Forage) | Jun 2025
- Foundations: Data, Data, Everywhere — Google / Coursera | Jun 2025

EDUCATION

B.Tech – Computer Science and Engineering
B.S. Abdur Rahman Crescent Institute of Science & Technology, Chennai | 2022 – 2026

Higher Secondary (Class XII)
Oxford Matric Higher Secondary School, Thiruvarur | 2020 – 2022

LANGUAGES

English — Professional Proficiency Tamil — Native